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Esercizio 424.

$$y = \frac{1}{1+2x}$$

$$\frac{1}{y} = 1 + 2x \implies \frac{1}{y} - 1 = 2x$$

$$\frac{1-y}{y} = 2x \implies x = \frac{1-y}{2y}$$

$$f^{-1}(x) = \frac{1-x}{2x}$$

Esercizio 425.

$$y = \sqrt[3]{x-3}$$

$$y^3 = x - 3 \implies x = y^3 + 3$$

$$f^{-1}(x) = y^3 + 3$$

Esercizio 426.

$$y = \sqrt{2x-4}$$

$$y^2 = 2x - 4 \implies y^2 + 4 = 2x$$

$$x = \frac{y^2 + 4}{2}$$

$$f^{-1}(x) = \frac{x^2 + 4}{2}$$

Esercizio 427.

$$y = 2^{x+3} + 4$$

$$y - 4 = 2^{x+3}$$

$$\log_2(y - 4) = x + 3$$

$$x = \log_2(y - 4) - 3$$

$$f^{-1}(x) = \log_2(x - 4) - 3$$

Esercizio 428.

$$y = -e^{-2x} - 1$$

$$1 - y = e^{-2x} \implies \ln(1 - y) = -2x$$

$$x = \frac{\ln(1 - y)}{-2}$$

$$f^{-1}(x) = \frac{\ln(1 - x)}{-x}$$

Esercizio 429.

$$y = 1 - 2 \ln(x - 2)$$

$$2 \ln(x - 2) = 1 - y \implies x - 2 = e^{\frac{1-y}{2}}$$

$$x = e^{\frac{1-y}{2}} + 2$$

$$f^{-1}(x) = e^{\frac{1-x}{2}} + 2$$

Esercizio 457.

$$f(x) = \frac{1}{4x^2}; \quad g(x) = 4x^2$$

$$f \circ g(x) = f(g(x)) = \frac{1}{4(4x^2)^2} = \frac{1}{64x^4}$$

$$g \circ f(x) = g(f(x)) = 4 \left(\frac{1}{4x^2} \right)^2 = \frac{1}{4x^4}$$

Esercizio 458.

$$f(x) = 2^x; \quad g(x) = \sqrt{x} - 2$$

$$f \circ g(x) = f(g(x)) = 2^{\sqrt{x}-2}$$

$$g \circ f(x) = g(f(x)) = \sqrt{2^x} - 2$$

Esercizio 459.

$$f(x) = \ln 2x; \quad g(x) = e^{-x}$$

$$f \circ g(x) = f(g(x)) = \ln(2(e^{-x})) = \ln(2e^{-x}) = \ln 2 + \ln(e^{-x}) = \ln 2 - x$$

$$g \circ f(x) = g(f(x)) = e^{-\ln 2x} = \frac{1}{2x}$$

Esercizio 460.

$$f(x) = \sin(2x); \quad g(x) = \sqrt{x} - 1$$

$$f \circ g(x) = f(g(x)) = \sin(2(\sqrt{x} - 1)) = \sin(2\sqrt{x} - 2)$$

$$g \circ f(x) = g(f(x)) = \sqrt{\sin 2x} - 1$$